



Department of Information Technology
College of Engineering, Guindy
Anna University, Chennai-25



LONG-TERM MEMORY ENGINE FOR AI AGENTS IN RUST

Second Project Review

GUIDE:

DR. BAMA SRINIVASAN

DEPARTMENT OF INFORMATION TECHNOLOGY

CEG, ANNA UNIVERSITY

TEAM MEMBERS:

M. KRITHIKRAJ, 2023115004

ALSTON S, 2023115006

ARYAN I, 2023115021

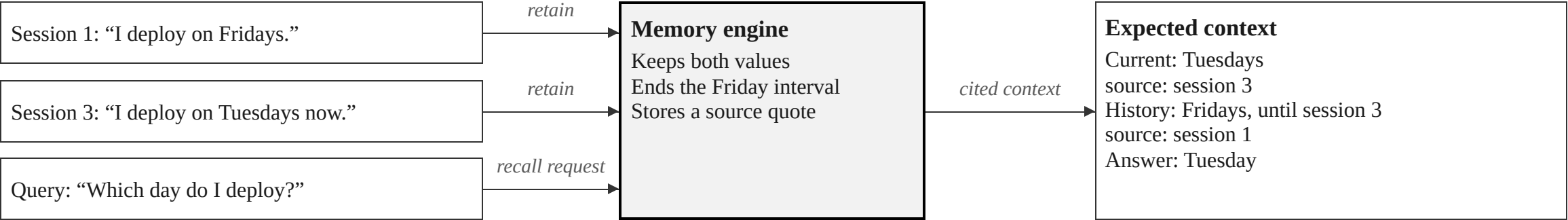
HARI SRINIVASAN, 2023115022

PROBLEM STATEMENT

1 WHY IT IS HARD

- Context limits. A full history costs a lot of tokens, and models tend to miss facts in the middle of a long input [1].
- Forgetting. A new session starts empty unless the old one was stored.
- Corrections. A new value should replace the old one without erasing it.
- Time. A question like “what did I decide last month” needs dated facts [2].
- Provenance. An answer should point to the evidence it came from.

3 WORKED EXAMPLE



2 ENGINEERING STATEMENT

Input: an append-only stream of interaction events (user turns, assistant turns, tool results), each with a timestamp.

Keep: the raw evidence, and every fact derived from it with a validity interval.

Return: at query time, a context within a token budget. Each item names its source evidence and says whether it is current.

LITERATURE SURVEY: MEMORY SYSTEMS

Work and source type	Mechanism	Evidence as reported by the source	Limitation	Relevance here
MemGPT, later Letta, Packer et al., 2023 [3]. Preprint.	Prompt split into main context and external recall and archival storage. The model moves data between them with function calls.	Authors’ own tests on OpenAI models. Deep memory retrieval: 92.5% with GPT-4 plus MemGPT against 32.1% for GPT-4 reading a summary.	Memory edits depend on the model’s function calling. No latency or token cost figures in the paper.	Origin of memory kept outside the prompt. Here, service code applies the version rules to claims the model extracts.
Zep and Graphiti, Rasmussen et al., 2025 [4]. Preprint, vendor-authored.	Temporal knowledge graph of episodes, entities and facts. Each fact has valid and invalid times, and a contradicting fact ends the old one.	Authors’ own LongMemEval run: 71.2% with gpt-4o against 60.2% for the full 115k-token history.	Single configuration, no repeated runs. The authors call their other benchmark (DMR) weak. Needs a graph database.	Closest design for validity intervals. We also run Neo4j, but only as a projection we can rebuild.
Mem0, Chhikara et al., 2025 [5]. Peer reviewed (ECAI 2025), authors are Mem0 staff.	An LLM extracts candidate facts, compares each with the nearest stored memories, then chooses add, update, delete or no-op.	LoCoMo only, authors’ run. Judge score 66.9 for Mem0, 72.9 for the full history, which had a p95 latency of 17 s against 1.4 s.	One benchmark and an unnamed judge model. The open-source release has since dropped update, delete and graph memory.	Same extract then compare idea. Here the update rule is code (cardinality and correction flag), not an LLM choice per fact.
Supermemory, Daga et al., 2026 [6]. Company report, not peer reviewed.	Atomic memories linked to source chunks, with updates, extends and derives links and two timestamps.	Reports LongMemEval-S results by category from its own run. Numbers not repeated here.	Engine source closed. The headline depends on the answer model, and the metric “aggregation” is not defined.	Named as a design influence in our repository. Used for mechanism only.
Hindsight, Latimer et al., 2025 [7]. Preprint, vendor-affiliated.	Retain, recall and reflect operations. Recall fuses four retrieval strategies by rank fusion and packs to a token budget.	Reports LongMemEval and LoCoMo results run by the authors. Numbers not repeated here.	Not peer reviewed. No ablation or variance found, and the token budget used is not stated.	Named design influence. We have no reranker.

CHALLENGES AND ISSUES IDENTIFIED

1 WHAT WE DID WRONG IN THE DESIGN

- We treated memory as plain chunks of text and searched them, with no context around each fact.
- Chunk search cannot tell that a new fact replaces an old one, so old and new values came back together.
- Nothing linked related facts, so a question that needs two connected facts could find only one of them.
- We had no way to show where an answer came from.
- Our first small distilled model was trained on the old memory format, so we retrained it for the fact format the graph uses.

2 WHAT WENT WRONG WHEN WE RAN IT

- Memory was per repository, so a session in another directory saw nothing.
- We re-sent the whole chat every turn: 4 turns stored 32 events and only 14 were different.
- A fresh session right after the first got no memory, because we embedded only after extraction.
- A bad model reply was cached, so every retry repeated it and later jobs got stuck behind it.
- The model re-extracted the old Friday value as new, and the current Tuesday value was lost.
- Recall had no distance limit, so about 2,000 tokens of unrelated text went in before almost every prompt.
- The date planner called a model on words like “when”, which took 4.46 s and 19.28 s on two queries.
- The default worker had no credentials, and Ollama silently cut 15 of 18 batches at its 4,096 token default.

3 WHAT THE KNOWLEDGE GRAPH FIXED

Facts are nodes with links (supersedes, contradicts, extends). A correction keeps the old fact as history. Every fact points to its exact quote and raw event, and recall follows links to related facts.

Prof. Vijaylaxmi Ma’am’s guidance pointed us to the knowledge graph.

PROJECT OBJECTIVES

Objective	Target	Measured	Status
Provenance	100% of stored memories pass an automated provenance check.	323 of 323 facts in user:dev have a source, and every quote is found word for word in its stored text.	Completed
Correct version history	Zero incorrect active versions on held-out update cases.	0 of 106 single-value slots hold two active facts, and the scripted Pi check (told, corrected, asked again) passes. There is no held-out update set yet, and only 2 facts are superseded.	In progress
Small local extractor	Student F1 within 5 points of the teacher at one third of its latency.	Retrained as Qwen3-4B and served by Ollama as the worker. Internal test: consolidation 1 of 16 to 13 of 16, reflection 10 of 16 to 15 of 16, but extraction coverage fell from 50 of 55 to 41 of 55. The 0.398 F1 was for the old 1.7B model.	In progress
Retrieval latency	p95 retrieval at most 300 ms at 10 QPS.	Date planning off: p50 58 ms, p95 72 ms, p99 75 ms (300 requests, 0 errors). With the date planner in the path p95 is 1,870 ms, because 4 of 40 test questions start a model call of 1.6 to 2.1 s.	In progress
Context size	Median returned context at most 1,000 tokens.	Median 1,942 tokens, p95 2,041, with the 2,048 token budget Pi sends (characters divided by four). Target not met.	In progress
Reproducibility	Models, prompts and configuration pinned.	Worker model pinned to a Hugging Face revision and SHA-256 checksum. The Docker image tag still floats.	In progress
Cross-session memory in Pi (new)	A fact told in one Pi session is recalled in a fresh session in another directory.	14 of 14 scripted checks in each of 4 runs (claude-haiku-4-5 or claude-sonnet-5-5 through Pi). One scenario, one machine.	Completed

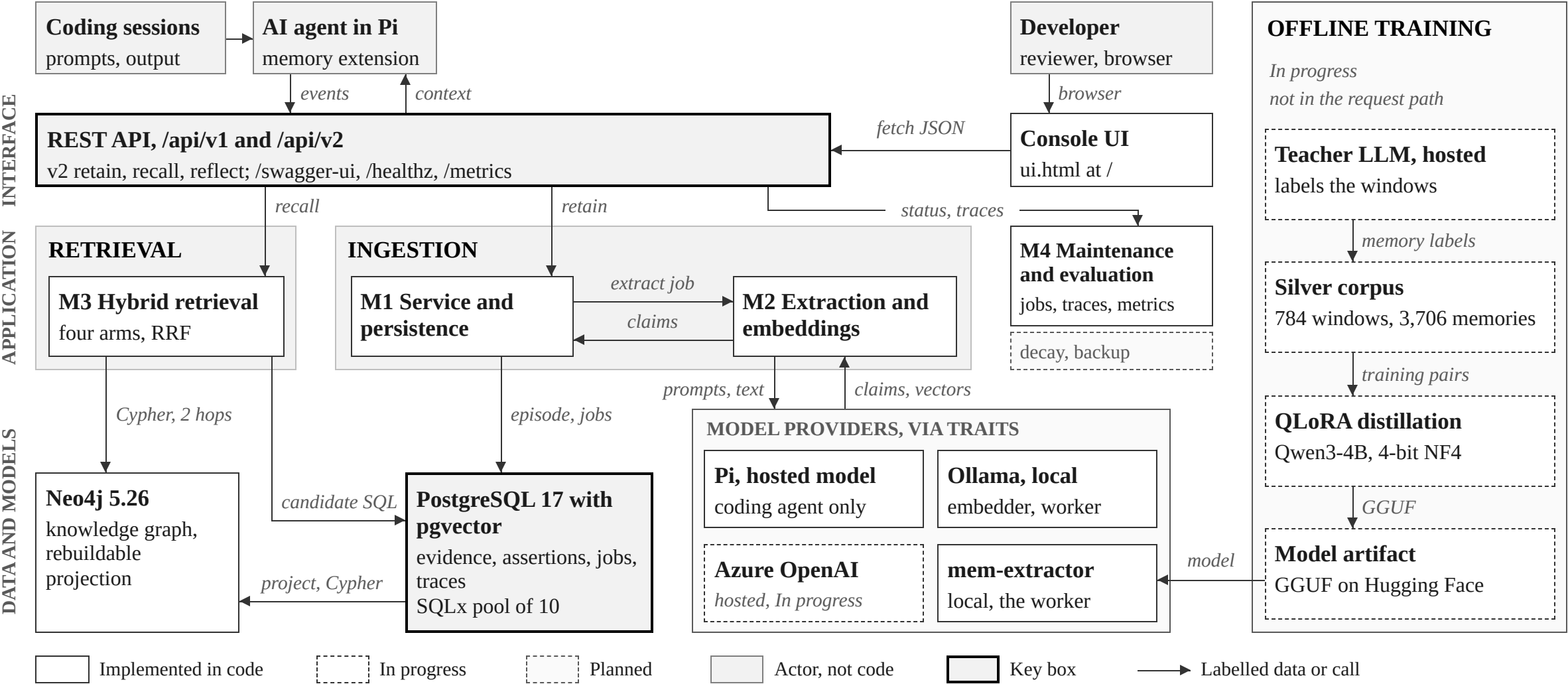
CHANGES SINCE FIRST REVIEW

Area	First Review	Now
Stores	One PostgreSQL 17 database, no graph store	PostgreSQL 17 with pgvector, plus a Neo4j 5.26 graph projection
API	Four routes: session events, request status, context, delete	/api/v1 with 8 routes and /api/v2 with 13 (retain, recall, reflect, jobs, graph)
Write path	One transaction, then 202 and a request id	Retain stores episode, chunks and an extract job in one transaction. Workers extract later.
Extraction	Distilled Qwen3-1.7B student on the serving path	Fine-tuned Qwen3-4B worker (mem-extractor) served by Ollama, with exact-quote checks and two repairs. Pi is only the coding agent.
Retrieval	Full-text and vector search, rank fusion	Lexical, semantic, temporal and graph searches, rank fusion (k = 60), raw evidence fallback, optional distance cutoff
Agent integration	None	Pi extension: spool, retain, recall, injected context
Graph view	None	Live page at /graph that draws the memory of one namespace from the real data
Tests	None reported	Rust 83, Python 5 and TypeScript 9, all passing (slide 15)
Observability	Request traces, OpenTelemetry	Traces in PostgreSQL, /healthz, /metrics, JSON logs

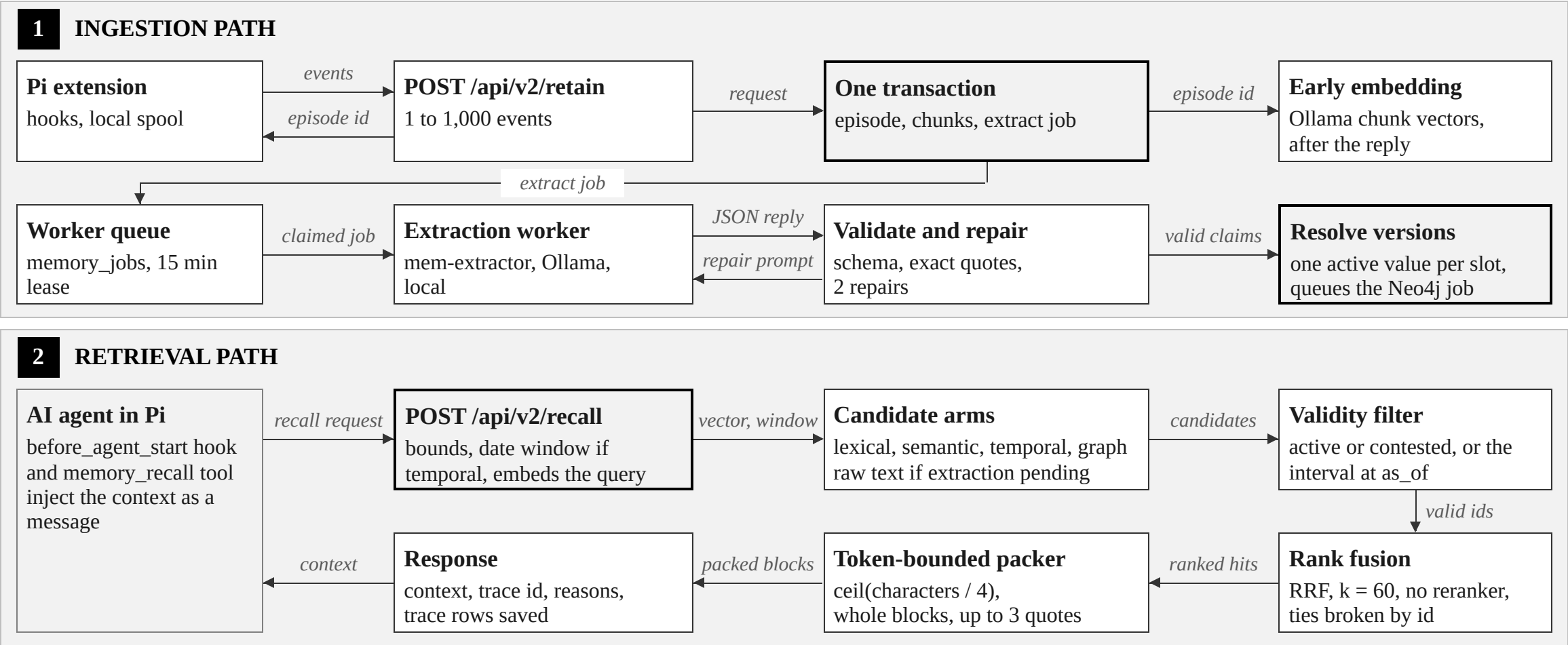
Claims that no longer hold

- First Review: no second store. Now: PostgreSQL plus a Neo4j 5.26 graph projection.
- First Review: the student is Qwen3-1.7B. Now: it is Qwen3-4B, retrained on the fact format and wired in as the worker.
- First Review: no memory data leaves the machine. Now: extraction runs locally, but Pi sends the coding chat to its own hosted model.

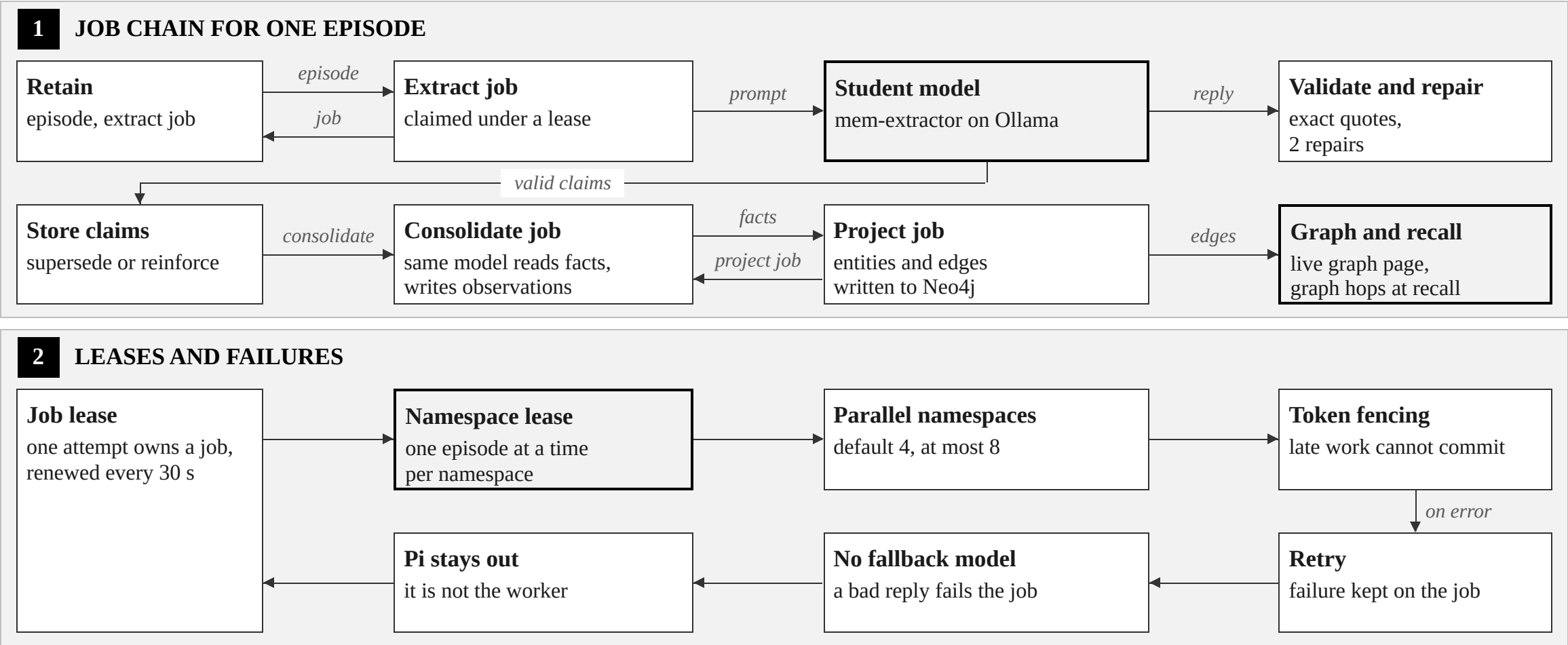
REFINED SYSTEM ARCHITECTURE



INGESTION AND RETRIEVAL PATHS



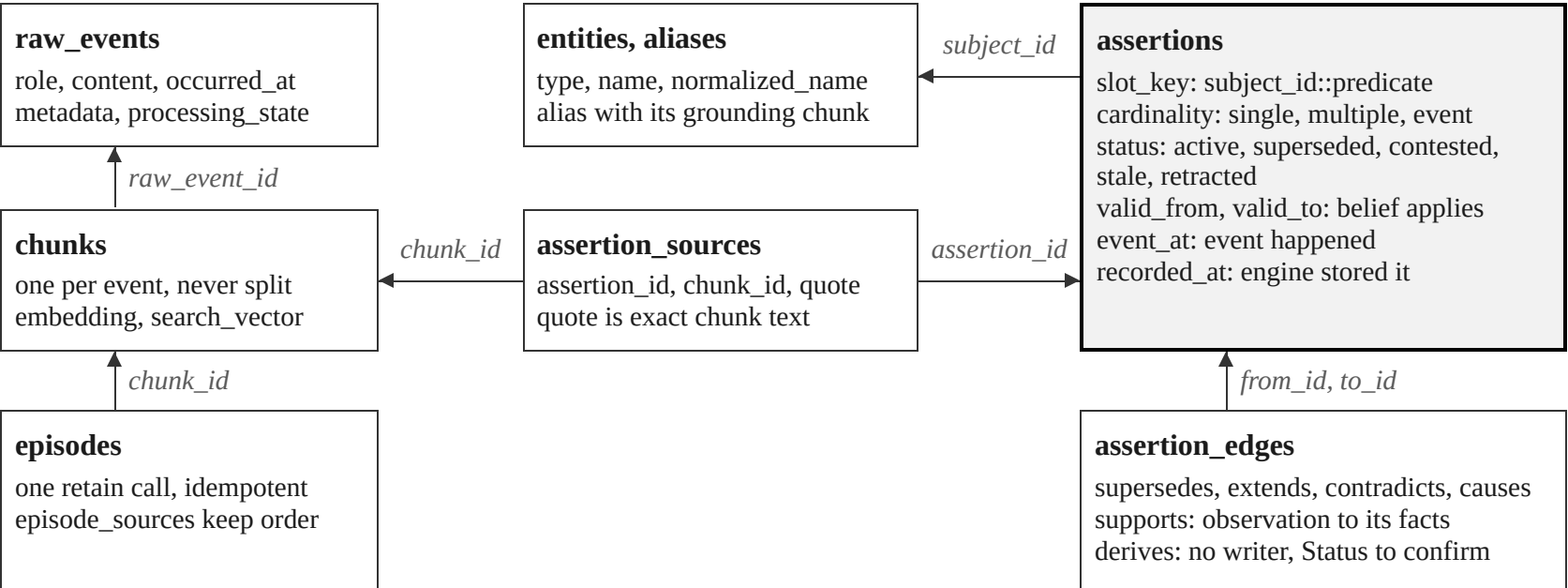
WORKER ARCHITECTURE



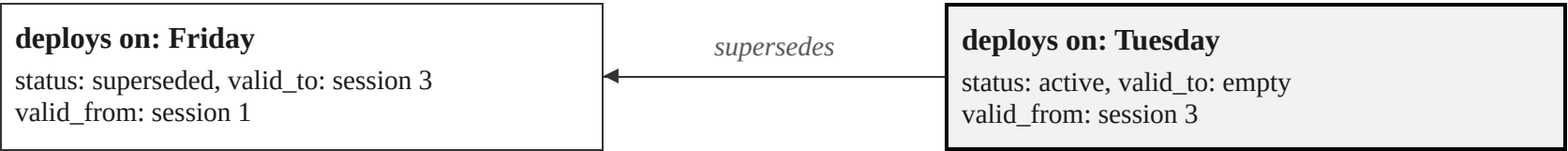
 Implemented in code  Actor, not code  Key box  Labelled data or call

DATA MODEL AND KNOWLEDGE GRAPH

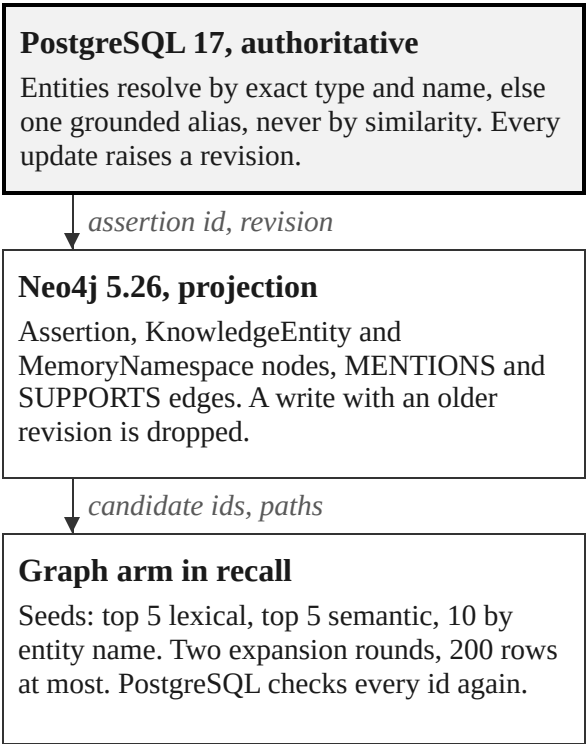
1 TABLES AND LINKS



2 VERSION CHAIN FOR ONE SLOT

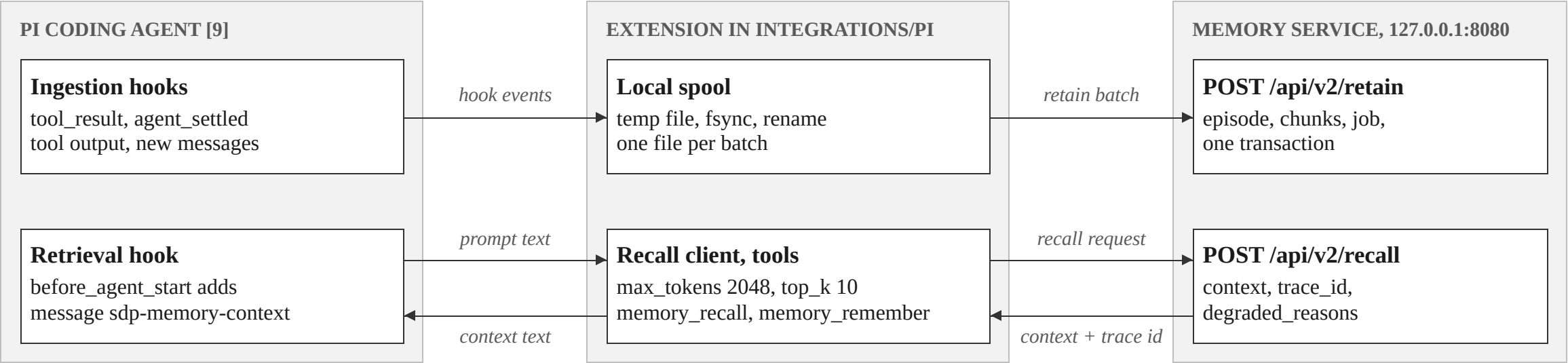


3 GRAPH PROJECTION



Every fact traces back to its exact quote, chunk and raw event. Graph results are ranked by Neo4j row order, and we have not measured what the graph search adds.

PI INTEGRATION



What	How it works
Namespace	One per user, user:<login>, shared by all directories
Retention	New messages only, tracked by a cursor entry
Automatic recall	Date planner off, 15 s timeout, raw text included, max_distance 0.45
Tests	Completed, 9 TypeScript tests

Recall cutoff: Completed
Automatic recall passes max_distance 0.45. The memory_recall tool passes none.

INSTALL, PRIVACY AND CHECKS

- Not decided yet: a global install with pi install, and whether every session may be retained.
- Not retained: the injected message and the results of the memory tools.
- Command /memory-status flushes the spool and shows job state.
- Scripted check, 14 checks per run: 14 of 14 in four runs (claude-haiku-4-5 or claude-sonnet-5-5 through Pi). One fact, one machine.

ALGORITHMS

1 WRITE PATH: HOW A CONVERSATION BECOMES FACTS

```
extract(episode):
prompt = rules + 80 newest current facts + events
for attempt in 1..3: (call, then 2 repairs)
claims = model(prompt)
error = validate(claims) (schema, quotes, ids)
if error: prompt += claims + error, else stop
if error: fail the job (backoff, 5 attempts)
embed missing chunks, then claim statements
for claim in claims: (one transaction)
slot = subject_id :: predicate
start = claim.valid_from, else first cited event time
same slot and value: add quote, keep status
multiple, event, or no old row: insert as active
single, correction, confidence >= 0.8:
old: superseded, valid_to = max(old.valid_from, start)
new: active, edge new supersedes old
single otherwise: old and new become contested
old row left active: mark dependants stale
consolidate job: observation needs 2+ active supports
```

Rank fusion (RRF), in words

$\text{score}(d) = \text{sum over searches } i \text{ of } 1 / (60 + \text{rank}_i(d))$ [12]
Each search ranks its results. A result gets $1 / (60 + \text{rank})$ from every search that found it, and the scores add up. The 60 is the paper's constant, not tuned.

2 READ PATH: HOW A QUESTION BECOMES CONTEXT

```
recall(query, as_of, max_tokens = 8192, top_k = 20):
if time words in query: ask model for a date window
lexical = full text search (valid rows, 40)
semantic = nearest(embed(query)) (valid rows, 40)
temporal = newest in window (if a window is set)
graph = expand(seeds), keep valid ids
if max_distance: drop far vector hits
if all arms empty, or extraction pending, or include_raw:
add raw chunk matches
score each id by RRF, sort by score, then by id
used = 0, blocks = 0
for block in ranked:
if blocks == top_k: stop
if used + tokens(block) <= max_tokens:
add block, used += tokens(block), blocks += 1
return context, trace_id
# In words: four searches each rank the facts they find, the ranks are
# fused into one list, and the best blocks that fit the budget are returned
```

Token estimate

$\text{tokens}(\text{block}) = \text{ceil}(\text{characters} / 4)$
A rough rule for English text. It stops us adding blocks once the budget is full.

What validation proves

Each quote occurs word for word in the event it cites. It may still not support the claim.

TOOLS AND DATASET

Tool	Version	Actual role	Runs
Rust, Tokio, Axum, SQLx	1.96.1, 1.53.1, 0.8.9, 0.9.0	Service, async runtime, REST API and database access	Local
PostgreSQL, pgvector	17, pgvector 0.8.6	Authoritative store, full-text search [10] and HNSW cosine index [11]. Compose pins the floating tag only	Local, Docker
Neo4j	5.26	Graph projection over the HTTP endpoint, marked deprecated in 5.26 [8]	Local, Docker
Ollama	0.13.5	Qwen3-Embedding-0.6B, 1024-d [13], and the mem-extractor worker: Qwen3-4B, Q8_0 GGUF, 4.3 GB, pinned Hugging Face revision	Local
Pi coding agent	1.0.2 locked, 0.85.0 installed	Hosts the memory extension [9]. It is the coding agent, not the extraction worker. Completed	Local agent, hosted model
Azure OpenAI	text-embedding-3-small	Extractor and embedder behind the same traits. In progress: written, not in the main code yet	Hosted
Qwen3-4B student	4-bit NF4 QLoRA [14]	Fine-tuned for extraction, consolidation and reflection. Published as a GGUF and served by Ollama as mem-extractor	Local

Development and demo data

Namespace user:dev: the project’s own documents and source, 79 batches through POST /api/v2/retain, plus invented example facts from four Pi recordings.

Scripted Pi check: one fact told, corrected and asked from fresh sessions, 14 checks per run.

Extraction-training data

2,961 task rows in sixteen domains: 2,095 train, 424 validation, 442 test, from 1,181 episode IDs including replay. Labelled by a hosted teacher.

LoCoMo [15] is used only for evaluation, never for training. It is CC BY-NC 4.0.

OUTPUT SCREENSHOTS

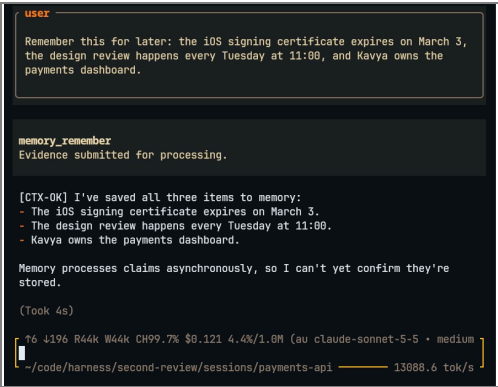


Figure 1. Session 1 in payments-api, 2026-10-06. The user tells Pi three example facts: the iOS signing certificate expires on March 3, the design review is every Tuesday at 11:00, and Kavya owns the payments dashboard. Pi calls memory_remember, which answers "Evidence submitted for processing", and replies that it saved all three items but cannot yet confirm they are stored. The CTX-OK marker comes from the presenter's own Pi instructions, not from the memory system.

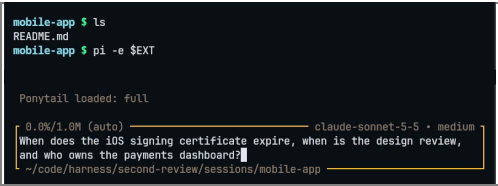


Figure 2. Session 2 in mobile-app, a fresh start with no history: ls shows only README.md. The question about the certificate date, the design review and the dashboard owner is typed in the editor. The empty lower part of the terminal is cropped away.

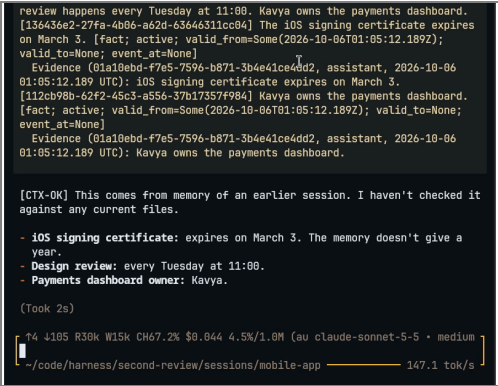


Figure 3. Session 2 after the answer. The upper part is the end of the memory the extension injected from session 1, so its first line starts mid-sentence: the facts, each with status active and an evidence line. Below it the answer gives March 3 (no year), every Tuesday at 11:00 and Kavya.

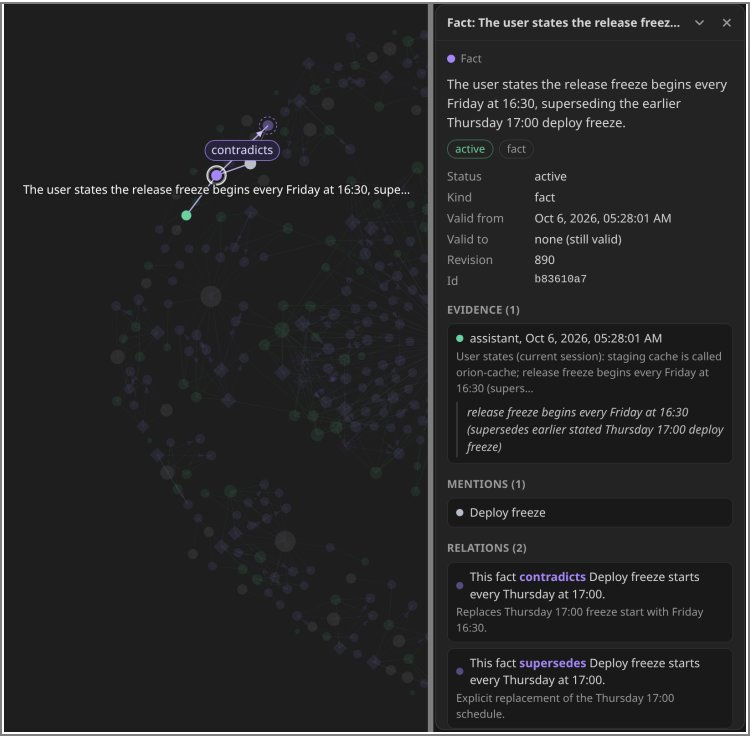


Figure 4. Graph page, user:dev, 2026-10-06. Two crops of one capture, joined at the grey line. Left: the selected fact (white ring), a contradicts relation and the superseded Thursday 17:00 fact (dashed ring). Right: its details panel, with status active, the evidence quote, the entity Deploy freeze and the relations contradicts and supersedes. Invented example facts.

RESULTS AND OBSERVATIONS

Measured finding	Result	Limit kept with it
Retrieval latency and context size	Date planning off: p50 58 ms, p95 72 ms, p99 75 ms at 10 requests per second (300 requests, 0 errors). With the planner: p95 1,870 ms, because 4 of 40 questions start a model call of 1.6 to 2.1 s. Median context 1,942 tokens, p95 2,041	One machine, local embeddings, 323 facts. Not tried on a large namespace. Tokens are characters divided by four
Provenance and version audit	323 of 323 facts have a source with a quote found word for word in its chunk. 0 of 106 single-value slots hold two active facts. Memory at idle: service 35 MB, PostgreSQL 224 MiB, Neo4j 1.27 GiB	Run on user:dev only, where just 2 facts are superseded, so it is not a held-out update set
Pi sessions, recording and scripted check	Session 2 (claude-sonnet-5-5), in another directory with no history, answered all three example facts. The scripted check passed 14 of 14 in four configurations	One scenario, invented facts, one machine, run by us. Shows the path works, not answer quality
Recall cutoff, 80 queries	Target fact in the context: 39 of 55 with no cutoff, 49 at 0.45. Follow-ups and small talk: mean context 6,925 to 1,250 characters	The same queries tuned the cutoff, so the claim is weak. Short prompts such as “ok” still pull in 3 to 5 kB
Local 14B extraction, 10 batches	Validation passed on 2 of 10 batches, the hosted model on 10 of 10. Before the context fix Ollama cut 15 of 18 prompts at 4,096 tokens, now 16,384 in and 4,096 out. GPU memory 10.6 GB at 4,096 and 12.9 GB at 16,384	One machine, one model, no repeats, stopped at 10 of 18 batches. Not usable for extraction yet
Live graph, user:dev	Neo4j projection: 382 nodes, 473 relationships. 133 of the 147 memories support at least one fact. Graph API counts equal direct Cypher and SQL counts, about 0.10 s per call at limit 200	One user, invented example facts. Chrome only, untested above a few thousand nodes
Automated tests	scripts/test.sh exits 0: Rust 83 (45 unit, 24 evidence_graph, 13 ollama_context, 1 postgres_api), Python 5, TypeScript 9	Database tests need TEST_DATABASE_URL and TEST_NEO4J_URL. Three live Ollama tests are opt-in and passed once
Fine-tuned worker, internal test	Consolidation 1 of 16 to 13 of 16, reflection 10 of 16 to 15 of 16. Extraction coverage fell from 50 of 55 to 41 of 55 facts	Labels and grading share a teacher family, so not independent human accuracy. Relative dates still wrong in places

STATUS AND PENDING WORK

Module	Implemented functionality	Available evidence	Pending work	Status
M1 Service and persistence	REST API (8 V1, 13 V2 routes), console, PostgreSQL store, job queue	src/api.rs, knowledge_store.rs; 83 Rust tests	Authentication and hard delete: Status to confirm	Completed
M2 Extraction and embeddings	JSON claims, exact-quote checks, two repairs; Ollama embeddings, 1024-d	src/worker.rs, providers.rs. Pi check: 14 of 14	Extraction quality at scale is unmeasured	Completed
Local 14B extraction	Qwen2.5-14B in Ollama, 16,384 token context	Passed validation on 2 of 10 test batches, hosted 10 of 10	Run the other 8 batches. Planned: constrained decoding, fewer related ids, a larger repair budget	In progress
M3 Hybrid retrieval and graph	Four searches, RRF k = 60, validity filter, token packing, Neo4j projection, live graph page, distance cutoff	src/v2.rs, graph.rs, tests/evidence_graph.rs; latency on slide 15	Shorter context, graph ablation, a rule for short prompts. A derives writer and a move off the deprecated Neo4j HTTP endpoint: Status to confirm	Completed
M4 Maintenance and evaluation	Retries, heartbeats, consolidation, traces, /healthz, /metrics	src/worker.rs	Human audit (Planned). Decay, re-embedding, backup: none exists	Completed
Pi integration	Hooks, local spool, memory tools, injected context message	integrations/pi/, 9 tests, scripted check 14 of 14	Global install and the privacy of retaining all sessions: Status to confirm	Completed
Fine-tuned worker (mem-extractor)	QLoRA of Qwen3-4B on 2,095 rows; Q8_0 GGUF served by Ollama	slm-distill/, release receipt	Coverage fell, relative dates weak. Live smoke check passed	In progress
Azure adapter and V1 console	AzureExtractor, AzureEmbedder, V1 console, browse and stats endpoints	Written: 12 files, 1,587 lines added	Rebase onto the V2 code (conflicts in src/main.rs and src/ui.html), then add a JsonModel adapter for the V2 worker	In progress

REFERENCES

- [1] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Trans. Assoc. Comput. Linguist.*, vol. 12, pp. 157-173, 2024. [Online]. Available: <https://aclanthology.org/2024.tacl-1.9/>
- [2] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025, pp. 86809-86836. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html. Also arXiv:2410.10813.
- [3] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," *arXiv preprint arXiv:2310.08560*, Oct. 2023 (rev. Feb. 2024). [Online]. Available: <https://arxiv.org/abs/2310.08560>
- [4] P. Rasmussen, P. Paliychuk, T. Beauvais, J. Ryan, and D. Chalef, "Zep: A temporal knowledge graph architecture for agent memory," *arXiv preprint arXiv:2501.13956 [cs.CL]*, Jan. 2025. [Online]. Available: <https://arxiv.org/abs/2501.13956>
- [5] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *Proc. 28th Eur. Conf. Artif. Intell. (ECAI)*, Oct. 2025, ser. *Frontiers in Artificial Intelligence and Applications*, vol. 413. IOS Press, 2025, pp. 2993-3000, doi: 10.3233/FAIA251160. Also arXiv:2504.19413 [cs.CL].
- [6] S. Daga, S. Sreedhar, and D. Shah, "LongMemEval," *supermemory research (vendor-hosted evaluation report on LongMemEval-S, not peer reviewed)*, May 2026. [Online]. Available: <https://supermemory.ai/research/longmembench/> (accessed Oct. 6, 2026).
- [7] C. Latimer, N. Boschi, A. Neeser, C. Bartholomew, G. Srivastava, X. Wang, and N. Ramakrishnan, "Hindsight is 20/20: Building agent memory that retains, recalls, and reflects," *arXiv preprint arXiv:2512.12818 [cs.CL]*, Dec. 2025. [Online]. Available: <https://arxiv.org/abs/2512.12818>
- [8] Neo4j, Inc., "Query the database," in *Neo4j HTTP API Manual*, Version 5 (page marked deprecated in 5.26). Neo4j, Inc., 2026. [Online]. Available: <https://neo4j.com/docs/http-api/5/query/> (accessed Oct. 6, 2026).
- [9] M. Zechner et al., "pi: AI agent toolkit (unified LLM API, agent loop, TUI, coding agent CLI)," package @earendil-works/pi-coding-agent, v1.0.2 (version locked by the project's package-lock.json; latest is 1.0.4 on Oct. 6, 2026), GitHub repository, 2026. [Online]. Available: <https://github.com/earendil-works/pi> (accessed Oct. 6, 2026).
- [10] PostgreSQL Global Development Group, "PostgreSQL 17 Documentation, Chapter 12: Full Text Search (Sections 12.2.2, 12.3.2, 12.3.3, 12.9) and Section 64.4: GIN Indexes," *PostgreSQL 17.11 documentation*, 2026. [Online]. Available: <https://www.postgresql.org/docs/17/textsearch.html> (accessed Oct. 6, 2026).
- [11] A. Kane et al., "pgvector: Open-source vector similarity search for Postgres," source code repository and README (0.8.6 installed in the test database, v0.8.7 tag also released), Oct. 2026. [Online]. Available: <https://github.com/pgvector/pgvector> (accessed Oct. 6, 2026).
- [12] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval (SIGIR '09)*, Boston, MA, USA, Jul. 2009, pp. 758-759, doi: 10.1145/1571941.1572114.
- [13] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, and J. Zhou, "Qwen3 Embedding: Advancing text embedding and reranking through foundation models," *arXiv preprint arXiv:2506.05176 [cs.CL]*, Jun. 2025. [Online]. Available: <https://arxiv.org/abs/2506.05176>
- [14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023, pp. 10088-10115, doi: 10.52202/075280-0441. Also arXiv:2305.14314.
- [15] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, Aug. 2024, pp. 13851-13870, doi: 10.18653/v1/2024.acl-long.747. [Online]. Available: <https://aclanthology.org/2024.acl-long.747/> (preprint: arXiv:2402.17753)